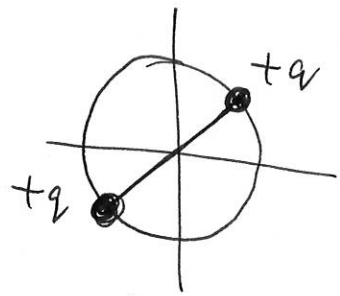
P1



$$Q_{\alpha\beta} = \sum_{j=1,2} \left(3\, r_{j\alpha} r_{j\beta} - \delta_{\alpha\beta}\, \vec{r}_j^{\,2}\right) q$$

$$= \left(-\delta_{\alpha\beta}\, 2R^2 + 2\cdot 3\, r_{1\alpha} r_{1\beta}\right) q$$

$$\vec{r}_1 = R(\cos\omega t, \sin\omega t)$$

$$\vec{r}_2 = -R(\cos\omega t, \sin\omega t)$$

$$Q_{xx} = 6R^2\cos^2\omega t - 2R^2 \quad , \quad Q_{xy} = 6R^2\cos\omega t\sin\omega t = Q_{yx}$$

$$Q_{yy} = 6R^2\sin^2\omega t - 2R^2$$

$$Q_{zz} = -2R^2$$

check that $\sum_\alpha Q_{\alpha\alpha} = 6R^2(\cos^2 + \sin^2) - 4R^2 - 2R^2 = 0$.

$$\Rightarrow \quad \hat{Q} = 2R^2 q \begin{pmatrix} 3\cos^2\omega t - 1 & 3\cos\omega t\sin\omega t & 0 \\ 3\cos\omega t\sin\omega t & 3\sin^2\omega t - 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\vec{Q}_\alpha = \hat{Q}_{\alpha\beta}\, n_\beta$$

$$Q_x = Q_{xx} n_x + Q_{xy} n_y = 2R^2 q\left[(3\cos^2\omega t - 1)\, n_x + 3\cos\omega t\sin\omega t\, n_y\right]$$

$$Q_y = Q_{yx} n_x + Q_{yy} n_y = 2R^2 q\left[(3\sin^2\omega t - 1)\, n_y + 3\cos\omega t\sin\omega t\, n_x\right]$$

$$Q_z = -2R^2 q\, n_z$$

$$\cancel{\vec{Q}\times\hat{n} = \hat{x}\quad \hat{y}\quad \hat{z}}$$

$$\cancel{3\cos^2 n_x}$$

$$Q_x = \left[6R^2\cos\omega t\,(\cos\omega t\, n_x + \sin\omega t\, n_y) - 2R^2 n_x\right] q$$

$$Q_y = \left[6R^2\sin\omega t\,(\cos\omega t\, n_x + \sin\omega t\, n_y) - 2R^2 n_y\right] q$$

$$Q_z = -2R^2 q\, n_z$$

Magnetic field due to quadrupole:

$$\vec{B} = \frac{\mu_0}{24\pi c^2 r} \dddot{\vec{Q}} \times \hat{n}$$

Since $\hat{n} = \text{const}$ ($\hat{n} = \hat{x}$ here), calculate $\vec{Q} \times \hat{n}$ first: use $\hat{n} = (1,0,0) = \hat{x} \Rightarrow n_y = n_z = 0$

$$\vec{Q} \times \hat{n} = \vec{Q} \times \hat{x} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ Q_x & Q_y & Q_z \\ 1 & 0 & 0 \end{vmatrix} = -Q_y \hat{z}$$

$$\vec{Q} \times \hat{n} = \left[-6R^2 q \sin(\omega t)\cos(\omega t)\right]\hat{z} = -3R^2 q\left[\sin(2\omega t)\right]\hat{z}.$$

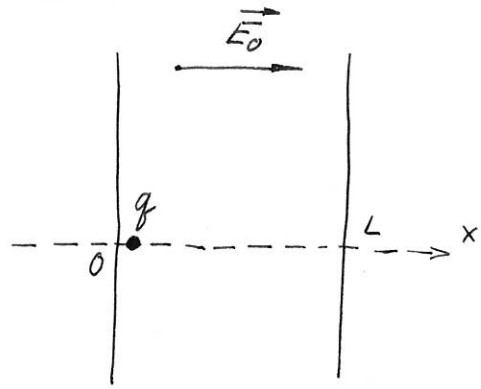
Thus $\dddot{\vec{Q}} \times \hat{n} = (-\hat{z}) q 3R^2 \frac{d^3}{dt^3}\left[\sin 2\omega t\right] = \hat{z}\, q 3R^2 (2\omega)^3 \cos(2\omega t)$

$$\vec{B} = \frac{\mu_0 q}{24\pi c^2 X_0} 3 \cdot 8 R^2 \omega^3 \cos(2\omega t)\hat{z} = \frac{\mu_0 q R^2 \omega^3}{\pi c^2 X_0}\cos(2\omega t)\hat{z}.$$

$$\vec{E} = c\vec{B} \times \hat{n} = \frac{\mu_0 R^2 \omega^3 q}{\pi c X_0}\cos(2\omega t)\hat{y}.$$

Remember that $t = t_r = t - \frac{X_0}{c}$ here is actually the retarded time.

Problem 2



Radiated power is given by Eq. (10.77):

$$P = \frac{\mu_0}{6\pi c} \ddot{P}_x^2 = \frac{\mu_0 q^2}{6\pi c} \ddot{x}^2$$

where acceleration is given by $\ddot{x} = \frac{qE_0}{m}$, so that $P = \frac{\mu_0 q^4 E_0^2}{6\pi c m^2}$

The time of travel is found from $L = \frac{qE_0}{m}\frac{t^2}{2}$

$$t = \sqrt{\frac{2Lm}{qE_0}}$$

The total radiated energy is thus

$$\Delta \mathcal{E}_{rad} = Pt = \frac{\sqrt{2}}{6\pi} \frac{\mu_0 q^{7/2} E_0^{3/2} L^{1/2}}{c m^{3/2}}$$

The energy acquired by the particle is simply

$$\mathcal{E}_{tot} = qE_0 L$$

The ratio of the two energies:

$$\frac{\Delta \mathcal{E}_{rad}}{\mathcal{E}_{tot}} = \frac{\sqrt{2}}{6\pi} \frac{\mu_0 q^{5/2} E_0^{1/2}}{c m^{3/2} L^{1/2}}$$

This ratio is small if L is large enough,

$$L \gg \frac{\mu_0 q^{5/2} E_0^{1/2}}{c m^{3/2} L^{1/2}}$$

Note that the ratio can be written as

$$\frac{\Delta \mathcal{E}_{rad}}{\mathcal{E}_{tot}} \sim \frac{\mathcal{E}_c \sqrt{\mathcal{E}_{tot}}}{\mathcal{E}_0^{3/2}}$$, where $\mathcal{E}_0 = mc^2$ is the

particle's rest energy,

$\mathcal{E}_C = \dfrac{q^2}{4\pi\varepsilon_0 L}$ is the Coulomb energy at distance L,

and

$\mathcal{E}_{tot}$ is defined above.

P3

$$m\ddot{\vec{r}} = \underbrace{-m\omega_0^2\vec{r}}_{\text{restoring force}} + \underbrace{q\vec{E}_0 e^{i\vec{k}\cdot\vec{r}-i\omega t}}_{\text{EM field}} + \underbrace{\frac{\mu_0 q^2}{6\pi c}\dddot{\vec{r}}}_{\text{radiation friction}}$$

↳ here we should, in principle, add $q\dot{\vec{r}}\times\vec{B}$ for Lorentz force, but for non-relativistic motion this represents a small correction ⇒ drop it.

Since amplitude of oscillations is much smaller than wave length, $\vec{k}\cdot\vec{r} \sim \frac{r}{\lambda} \ll 1$, and hence the exponential is approximated as $e^{i\vec{k}\cdot\vec{r}-i\omega t} \to e^{-i\omega t}$.

$$\ddot{\vec{r}} + \omega_0^2\vec{r} = \frac{q\vec{E}_0}{m}e^{-i\omega t} + \frac{\mu_0 q^2}{6\pi c m}\dddot{\vec{r}}$$

It is natural to try $\vec{r}(t) = \vec{r}_0 e^{-i\omega t}$ (driven solution)

$$(\omega_0^2-\omega^2)\vec{r}_0 e^{-i\omega t} = \frac{q\vec{E}_0}{m}e^{-i\omega t} + \frac{\mu_0 q^2}{6\pi c m} i\omega^3 \vec{r}_0 e^{-i\omega t}$$

$$\vec{r}_0 = \frac{q\vec{E}_0}{m\left[\omega_0^2-\omega^2-i\omega\gamma(\omega)\right]} \quad \text{where } \gamma(\omega) = \frac{\mu_0 q^2\omega^2}{6\pi c m}.$$

Then $\vec{d}(t) = q\vec{r}(t) = \dfrac{q^2\vec{E}_0 e^{-i\omega t}}{m\left(\omega_0^2-\omega^2-i\omega\gamma(\omega)\right)} =$

$$= \frac{q^2\vec{E}_0}{m}\,\frac{\left(\omega_0^2-\omega^2+i\omega\gamma(\omega)\right)\left(\cos\omega t - i\sin\omega t\right)}{\left(\omega_0^2-\omega^2\right)^2+\omega^2\gamma^2(\omega)} \quad (*)$$

But now we need to remember that $\vec{d}(t)$ is real variable (as any physical variable is) and therefore physical $\vec{d}$ is obtained by taking <u>real</u> part of $(*)$.

$$\vec{d}(t) = \frac{q^2 \vec{E}_0}{m} \frac{(\omega_0^2-\omega^2)\cos\omega t + \omega\gamma(\omega)\sin\omega t}{(\omega_0^2-\omega^2)^2 + \omega^2\gamma^2(\omega)}.$$

$$\ddot{\vec{d}} = -\frac{q^2 \vec{E}_0 \omega^2}{m} \frac{(\omega_0^2-\omega^2)\cos\omega t + \omega\gamma(\omega)\sin(\omega t)}{(\omega_0^2-\omega^2)^2 + \omega^2\gamma^2(\omega)}$$

Emitted radiation

$$\frac{dP}{d\Omega} = \frac{\mu_0}{16\pi^2 c}\left(\ddot{\vec{d}} \times \hat{n}\right)^2 = \frac{\mu_0}{16\pi^2 c}\left(\ddot{\vec{d}}\right)^2 \sin^2\theta$$

To average over time we need to remember (or just calculate)

$$\frac{1}{T}\int_0^T dt\, \cos^2(\omega t) = \frac{1}{T}\int_0^T dt\, \sin^2(\omega t) = \frac{1}{2} \quad \text{while} \int \cos\omega t \cdot \sin\omega t \to 0$$

Thus
$$\overline{\frac{dP}{d\Omega}} = \frac{\mu_0}{16\pi^2 c}\left(\frac{q^2 E_0}{m}\right)^2 \frac{\omega^4\left[(\omega_0^2-\omega^2)^2 + \omega^2\gamma^2(\omega)\right]}{\left[(\omega_0^2-\omega^2)^2 + \omega^2\gamma^2(\omega)\right]^2}\sin^2\theta$$

$$= \frac{\mu_0}{16\pi^2 c}\left(\frac{q^2 E_0}{m}\right)^2 \frac{\omega^4}{(\omega_0^2-\omega^2)^2 + \omega^2\gamma^2(\omega)}\sin^2\theta$$

Therefore total power, integrated over $d\Omega$,

$$\overline{P} = \frac{\mu_0}{16\pi^2 c}\frac{q^4 E_0^2}{m^2}\frac{\omega^4}{(\omega_0^2-\omega^2)^2 + \gamma^2(\omega)\omega^2}\underbrace{\int d\Omega \sin^2\theta}_{\frac{2}{3}\cdot 4\pi}$$

$$= \frac{\mu_0}{\cancel{16}\,8\pi c}\left(\frac{q^2 E_0^2}{m}\right)^2 \frac{\omega^4}{(\omega_0^2-\omega^2)^2 + \omega^2\gamma^2(\omega)} = S \cdot \frac{8\pi}{3}\left(\frac{q^2}{4\pi m \varepsilon_0 c^2}\right)^2 \frac{\omega^4}{(\omega_0^2-\omega^2)^2 + \omega^2\gamma^2(\omega)}$$

Intensity of incoming wave $S = c \cdot \varepsilon_0 \vec{E}^2 = \frac{1}{2} c\, \varepsilon_0 E_0^2$

↳ after time-averaging